

Rajalakshmi Engineering College

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2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 10_Q3

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : COD

1. Problem Statement

Priya is analyzing encrypted messages in a research project. She wants to analyze the frequency of each character in a given paragraph. The characters should be stored in a TreeMap so that the output is sorted in ascending order of characters automatically.

You are required to build a Java program that:

Uses a `TreeMap<Character, Integer>` to count how many times each character appears in the message. Ignores spaces and considers only alphabets (case-sensitive). Outputs the frequencies of characters in sorted order.

You must use a TreeMap in the class named MessageAnalyzer.

Input Format

The first line of input contains an integer n, the number of lines in the message.

The next n lines each contain a string (the encrypted message line).

Output Format

The first line of output prints: "Character Frequency:"

Then print each character and its frequency in the format: "<character>: <count>"

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 2
Hello World
Java

Output: Character Frequency:

H: 1

J: 1

W: 1

a: 2

d: 1

e: 1

l: 3

o: 2

r: 1

v: 1

Answer

```
import java.util.*;
```

```
class MessageAnalyzer{  
    Map<Character, Integer> tmap = new TreeMap<>();  
    private String words="";
```

```
    public void putString(String str){  
        words = words.concat(str);  
    }
```

```
    public void findFrequency(){
```

```

        for(char c: words.toCharArray()){
            tmap.put(c, tmap.getOrDefault(c, 0)+1 );
        }
    }

    public void displayFrequency(){
        System.out.println("Character Frequency: ");
        tmap.forEach((k,v)-> System.out.println(k+": "+v));
    }
}

public class Main{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();sc.nextLine();

        MessageAnalyzer msg = new MessageAnalyzer();

        for(int i=0; i<n; i++){
            String str = sc.nextLine();
            String[] parts = str.split(" ");
            for(String s: parts) msg.putString(s);
        }

        msg.findFrequency();
        msg.displayFrequency();
        sc.close();
    }
}

```

Status : Correct

Marks : 10/10